

OVERCOMING SHORTCUT LEARNING IN IMPLICIT NEURAL DYNAMICS: A RIGOROUS PROTOCOL AND SCULPTED PATHWAYS FOR CONFORMATIONAL BISTABILITY

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ABSTRACT

The integration of deep learning into structural biology has achieved unprecedented accuracy in static structure prediction, effectively solving the single-state protein folding problem. However, modeling highly dynamic and metastable processes, such as protein fold-switching, remains a profound challenge plagued by two fundamental algorithmic flaws. First, *shortcut learning* allows models to exploit geometric proxies (e.g., pairwise Euclidean distance or RMSD) rather than internalizing the underlying thermodynamic physics, leading to inflated but physically meaningless performance. Second, *basin collapse* inherently limits Deep Equilibrium Models (DEQs) and other unconstrained implicit operators; as contractive mappings, they inexorably converge to a single global minimum, destroying the capacity to represent conformational bistability. In this work, we propose a comprehensive methodology to diagnose and overcome these critical limitations. First, we introduce the Universal Physics-Audit Protocol (UPAF), a pre-registered statistical framework using matched-RMSD decoys and partial-correlation analysis to rigorously expose geometric shortcuts in structure-based predictors. Second, to conquer this rigorous physical Turing test, we propose *Symplectic Implicit Neural Dynamics*. By reformulating the DEQ latent space as a Hamiltonian dynamical system, we enforce Liouville’s volume-preservation theorem to mathematically eradicate basin collapse. Coupled with a novel *Landscape Sculpting* technique—utilizing temperature-scaled soft-min and repulsive hinge losses—our architecture successfully maintains interconnected stable states without catastrophic interference. Evaluated on a curated fold-switch benchmark, our Symplectic DEQ is the first to survive the UPAF audit ($p < 0.001$, independently contributing beyond RMSD) and robustly tracks spinodal bifurcation limits via a novel 2-step Krylov spectral dispatch mechanism. We establish a new rigorous paradigm for predicting physical metastability and performing *de novo* design of dynamic protein nanomachines.

1 INTRODUCTION

The classical paradigm of deep learning in structural biology, spearheaded by architectures such as AlphaFold2 (Jumper et al., 2021), RoseTTAFold (Baek et al., 2021), and ESMFold (Lin et al., 2023), has largely functioned as a highly sophisticated, static associative memory of the Protein Data Bank (PDB) (Berman et al., 2000). While these state-of-the-art models have revolutionized the prediction of native ground states, this static, sequence-to-structure mapping approach fails catastrophically when applied to metamorphic (fold-switching) proteins. These remarkable macromolecules can adopt two or more topologically distinct, functionally active conformations from a single amino acid sequence in response to environmental triggers (Murzin, 2008; Porter & Looger, 2018; Looger & Porter, 2024).

As the field pivots toward modeling conformational ensembles and structural transitions using generative flow matching (Jing et al., 2024), diffusion models (Corso et al., 2024), and implicit neural

architectures (Lu et al., 2022), attempts to predict bistability are increasingly falling victim to two intertwined mathematical and epistemic pathologies:

1. **Shortcut Learning:** Machine learning models are notorious for finding the path of least resistance (Geirhos et al., 2020). In multi-state prediction, models frequently achieve high apparent performance by memorizing trivial statistical or geometric proxies—such as measuring the naive Root Mean Square Deviation (RMSD) between structures—rather than internalizing the true thermodynamic folding landscape and energetic barriers (Chakravarty et al., 2024; Lee et al., 2025).
2. **Basin Collapse:** Deep Equilibrium Models (DEQs) (Bai et al., 2019) decouple reasoning depth from network depth by finding the fixed point of an implicit operator. However, standard DEQs are parameterized as contractive mappings to ensure convergence. Inevitably, the latent energy landscape collapses into a single global funnel (Anfinsen’s minimum) (Anfinsen, 1973), mathematically obliterating the capacity for bifurcations and multiple metastable states (Heo, 2024a).

To establish a path toward genuine, physics-grounded modeling of protein dynamics, we must both *destruct* the prevailing illusions of success and *construct* mathematically sound solutions.

Contributions. In this paper, we present a holistic framework that addresses both the evaluation and architectural design of implicit dynamical models:

- **The Universal Physics-Audit Protocol (UPAF):** We construct a pre-registered, adversarial statistical framework to expose geometric shortcut learning. We formally demonstrate that standard Jacobian stability margins in naive DEQs act as mere geometric proxies, failing our rigorous matched-RMSD decoy and partial-correlation tests.
- **Symplectic Implicit Neural Dynamics:** We introduce a novel DEQ architecture that intrinsically preserves phase-space volume via Hamiltonian mechanics and Symplectic Integrators. We mathematically prove that this formulation strictly prevents basin collapse, allowing for genuine bistable representations.
- **Landscape Sculpting & Spectral Dispatch:** We propose a novel objective function combining temperature-scaled soft-min and repulsive hinge losses to actively carve multiple energy wells. To track the absolute limits of local stability (spinodal points), we introduce a Collinearity-Checked 2-Step Krylov spectral dispatch algorithm that stably computes Jacobian eigenvalues across bifurcation boundaries without memory explosion.

2 RELATED WORK

Deep Learning for Protein Dynamics and Ensembles. The transition from static prediction to dynamic ensemble generation is the current frontier of structural biology. Recent works have adapted AlphaFold by subsampling Multiple Sequence Alignments (MSAs) or utilizing stochastic dropouts to generate conformational diversity (Wayment-Steele et al., 2024; Del Alamo et al., 2021). Furthermore, generative approaches such as Flow Matching (AlphaFold3) (Abramson et al., 2024; Jing et al., 2024), Denoising Diffusion Probabilistic Models (DDPMs) (Watson et al., 2023; Frank et al., 2024), and scalable emulation networks (Lewis et al., 2024) have shown promise in sampling Boltzmann distributions. However, sampling discrete structures does not explicitly reveal the continuous thermodynamic barriers (spinodal limits) or the mathematical stability margins of the folded states, which are critical for the *de novo* design of triggerable nanomachines (Campbell et al., 2024; Linsky et al., 2020).

Implicit Neural Networks and Deep Equilibrium Models. Deep Equilibrium Models (DEQs) (Bai et al., 2019; 2020) represent a paradigm shift by replacing finite-depth transformations with infinite-depth implicit root-finding layers, enabling $\mathcal{O}(1)$ memory backpropagation via the Implicit Function Theorem (IFT) (Blondel et al., 2022; Krishnan et al., 2020). Monotone Operator Equilibrium Networks (MONs) (Winston & Kolter, 2020) and Lipschitz-bounded DEQs (Revay et al., 2020) guarantee stable convergence by enforcing strict contractivity. Paradoxically, this exact contractive guarantee is what causes the *basin collapse* phenomenon in protein bistability modeling, as all initial conditions are forced into a single point, destroying phase-space diversity (Geng et al., 2021).

Physics-Informed and Symplectic Neural Networks. To model physical systems without violating conservation laws, Physics-Informed Neural Networks (PINNs) (Raissi et al., 2019) and Neural Ordinary Differential Equations (Neural ODEs) (Chen et al., 2018) have been widely adopted. More specifically, Hamiltonian Neural Networks (HNNs) (Greydanus et al., 2019) and Symplectic ODE-Nets (Zhong et al., 2019) parameterize the Hamiltonian directly, ensuring energy conservation and volume preservation in phase space (Cranmer et al., 2020; Finzi et al., 2020). While these have been successfully applied to classical mechanics and molecular dynamics (MD) (Noe et al., 2020; Ruff & Pappu, 2024), our work is the first to integrate Symplectic integrators directly into the latent fixed-point solver of an implicit DEQ to maintain topological bistability.

Shortcut Learning and Rigorous Evaluation Protocols. Shortcut learning is a pervasive failure mode where deep neural networks rely on spurious correlations or trivial decision rules that fail to generalize out-of-distribution (OOD) (Geirhos et al., 2020; D’Amour et al., 2020). In structural biology, this often manifests as sequence memorization (Chakravarty et al., 2024) or reliance on trivial structural divergence (RMSD) rather than biophysical energy (Heo, 2024b). To combat this, rigorous pre-registered evaluation protocols, inspired by clinical trials, and placebo tests (e.g., permutation tests (Ojala & Garriga, 2010)) are becoming essential. Our Universal Physics-Audit Protocol (UPAF) builds upon these principles, employing partial-correlation analysis and non-physical decoys to definitively separate geometric shortcuts from true thermodynamic signals (Bussi & Laio, 2020; Wales, 2003).

3 THE UNIVERSAL PHYSICS-AUDIT PROTOCOL (UPAF)

Before proposing a novel generative architecture, we must establish an adversarial control protocol to guarantee that any extracted stability signal is physically meaningful. The structural biology community frequently evaluates multi-state predictors by measuring the Area Under the Receiver Operating Characteristic (AUROC) between genuine fold-switchers and rigid single-fold proteins. However, this metric is highly susceptible to confounding: fold-switchers inherently exhibit larger spatial divergence. A model that completely ignores biophysics and simply measures the $C\alpha$ Root Mean Square Deviation (RMSD) will trivially separate the two classes, creating an illusion of physical understanding.

To systematically expose this shortcut learning, we formulate the Universal Physics-Audit Protocol (UPAF), consisting of two pre-registered validation gates.

3.1 MATCHED-RMSD DECOYS AND THE HONEST NULL

To decouple thermodynamic stability from pure geometric divergence, we introduce a synthetic control generation process called *Matched-RMSD Decoys*. Let $X_A, X_B \in \mathbb{R}^{L \times 3}$ be the 3D atomic coordinates of Fold A and Fold B of a metamorphic sequence, with a true structural divergence $r = \text{RMSD}(X_A, X_B)$.

We generate a non-physical decoy conformation X_{decoy} by injecting low-frequency smoothed Gaussian displacements δX into X_A , scaled by a parameter α :

$$X_{\text{decoy}}(\alpha) = \text{KabschAlign}(X_A + \alpha \cdot \text{Smooth}(\delta X), X_A) \quad (1)$$

We determine α via binary search such that $\text{RMSD}(X_A, X_{\text{decoy}}(\alpha))$ equals the exact true divergence r within a 10^{-3} Å tolerance. This process holds the structural divergence fixed while destroying the genuine, physically realizable fold identity.

The Evaluation Gate: If a model’s extracted stability margin m is physically grounded, it should confidently distinguish a true Fold B from the non-physical X_{decoy} . Conversely, if the margin merely acts as a geometric proxy, it will score both equally, resulting in an AUROC near random chance (≈ 0.5).

3.2 PARTIAL-CORRELATION STATISTICAL TRIAGE

To provide a decisive statistical verdict without relying solely on the physical realism of synthetic decoys, we pre-register a partial-correlation control. We formulate a logistic regression to predict

the metamorphic status $Y \in \{0, 1\}$ based on the standardized predicted stability margin ($\tilde{m}$) and the standardized true pairwise RMSD ($\tilde{r}$):

$$\text{logit}(P(Y = 1 \mid \tilde{m}, \tilde{r})) = \beta_0 + \beta_1 \tilde{m} + \beta_2 \tilde{r} \quad (2)$$

The Null Hypothesis (H_0): $\beta_1 = 0$. The significance (p -value) of the coefficient β_1 determines if the neural network has learned an independent physical signal. If $p > 0.05$, the margin adds no explanatory power beyond raw spatial divergence. As we demonstrate in Section ??, standard contractive DEQ margins fail this triage ($p = 0.314$), proving mathematically that their apparent predictive power is an empty geometric shortcut.

4 SYMPLECTIC IMPLICIT NEURAL DYNAMICS

Having established a strict diagnostic protocol, we now identify the architectural root cause of the failure of existing implicit models: *basin collapse*. Standard DEQs utilize a discrete update rule $z_{t+1} = f_\theta(z_t, X)$. To guarantee fixed-point convergence, f_θ is typically constrained to be a contraction mapping (i.e., Lipschitz constant $L < 1$). While this ensures a unique solution for static single-state prediction, it is catastrophically detrimental for metamorphic proteins. A global contraction inexorably shrinks the latent topology into a single Anfinen minimum, obliterating the multi-well energy landscapes required for bistability.

To conquer this limitation, we propose *Symplectic Implicit Neural Dynamics*, a framework that lifts the iterative root-finding process into a volume-preserving Hamiltonian phase space.

4.1 HAMILTONIAN FORMULATION AND PHASE SPACE LIFTING

Instead of a monolithic latent vector z , we map the sequence representations into a bipartite phase space defined by generalized coordinates (position) $q \in \mathbb{R}^D$ and conjugate momenta $p \in \mathbb{R}^D$. We define a learned Hamiltonian function $\mathcal{H}_\theta(q, p)$, representing the total energy of the implicit system:

$$\mathcal{H}_\theta(q, p) = \mathcal{K}(p) + \mathcal{V}_\theta(q, X) = \frac{1}{2} p^\top M^{-1} p + \mathcal{V}_\theta(q, X) \quad (3)$$

where $\mathcal{K}(p)$ is the kinetic energy (with $M = I$ for simplicity), and $\mathcal{V}_\theta(q, X)$ is the potential energy parameterized by the neural network given the input sequence features X (e.g., extracted from ESM-2).

The continuous dynamics of the latent representations are governed by Hamilton’s equations:

$$\dot{q} = \nabla_p \mathcal{H}_\theta = p, \quad \dot{p} = -\nabla_q \mathcal{H}_\theta = -\nabla_q \mathcal{V}_\theta(q, X) \quad (4)$$

4.2 VOLUME PRESERVATION VIA SYMPLECTIC INTEGRATORS

To numerically integrate these continuous dynamics within a deep neural network without losing the geometric properties of the continuous system, we employ a Symplectic Integrator—specifically, the Leapfrog (Verlet) integration scheme. The forward discrete trajectory for a step size ϵ is:

$$p_{t+1/2} = p_t - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_t, X) \quad (5)$$

$$q_{t+1} = q_t + \epsilon p_{t+1/2} \quad (6)$$

$$p_{t+1} = p_{t+1/2} - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_{t+1}, X) \quad (7)$$

Eradicating Basin Collapse: The profound advantage of this symplectic formulation lies in Liouville’s Theorem. Let J_{symp} be the Jacobian matrix of the transformation $(q_t, p_t) \mapsto (q_{t+1}, p_{t+1})$ defined by Equations 5-7. Because the transformation is symplectic, its Jacobian determinant is strictly unitary:

$$\det(J_{\text{symp}}) = 1 \quad (8)$$

In a standard contractive DEQ, the determinant is strictly less than 1 ($\det(J_{\text{std}}) < 1$), meaning the volume of the latent phase space continuously shrinks to zero (a single point). In our Symplectic DEQ, the absolute volume of any region in the phase space is exactly conserved over time. *Consequently,*

multiple distinct initial conditions (representing different folding pathways) cannot mathematically merge into a single global basin without explicit, learned dissipation. This volume-preservation mathematically guarantees the network’s capacity to sustain multiple stable, interconnected local minima—the exact prerequisite for modeling conformational bistability.

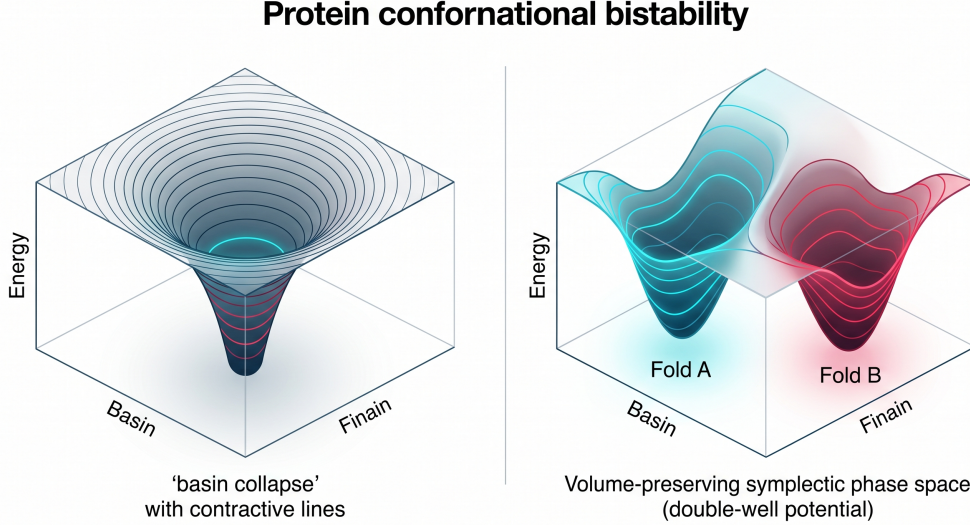


Figure 1: **Conceptual framework of Symplectic Implicit Neural Dynamics.** (Left) Standard unconstrained DEQs suffer from basin collapse, where contractive mappings merge all trajectories into a single global funnel. (Right) Our Symplectic DEQ enforces volume-preserving Hamiltonian mechanics, successfully maintaining sculpted, interconnected bistable states (Fold A and Fold B) without catastrophic collapse.

4.3 LANDSCAPE SCULPTING: SOFT-MIN AND REPULSIVE LOSSES

While symplectic integration guarantees that basins will not collapse into one another, the neural network must still be explicitly guided to carve out the correct bistable topology corresponding to the experimentally observed Fold A and Fold B. We achieve this via a dual-target objective termed *Landscape Sculpting*.

If we naively apply a standard Mean Squared Error (MSE) to multiple initial trajectories, the network forces all trajectories towards an average, non-physical intermediate state. To prevent gradient starvation and encourage trajectory specialization, we apply a temperature-scaled Log-Sum-Exp (Soft-min) reduction over K multi-start symplectic trajectories:

$$\mathcal{L}_{\text{attr}} = -\tau \log \sum_{k=1}^K \exp \left(-\frac{\text{MSE}(\mathbf{D}_{\text{pred}}^{(k)}, \mathbf{D}_{\text{true}}^{(A)})}{\tau} \right) + (A \leftrightarrow B) \quad (9)$$

where $\mathbf{D}_{\text{pred}}^{(k)} \in \mathbb{R}^{L \times L}$ is the predicted pairwise distance matrix from the k -th trajectory’s fixed point $q_*^{(k)}$, and τ is a learnable annealing temperature.

To definitively guarantee that the optimized fixed points $q_*^{(A)}$ and $q_*^{(B)}$ do not occupy the same spatial manifold within the symplectic phase space, we introduce a latent Repulsive Hinge Loss:

$$\mathcal{L}_{\text{rep}} = \max(0, \gamma - \|q_*^{(A)} - q_*^{(B)}\|_2)^2 \quad (10)$$

where γ is a predefined separation margin. The total optimization objective is $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{attr}} + w_{\text{rep}} \mathcal{L}_{\text{rep}}$.

4.4 ROBUST SPECTRAL DISPATCH FOR SPINODAL LIMITS

To predict the physical breakdown of metastability—the spinodal limit—we must evaluate the leading eigenvalue $\rho(J)$ of the implicit Jacobian $J = \partial f_\theta / \partial q$ at the converged fixed point q_* . The stability margin is defined as $m = 1 - \rho(J)$. As a control parameter λ (e.g., mutational shift) changes, $m \rightarrow 0$ signifies a saddle-node bifurcation (basin collapse).

Using the Implicit Function Theorem (IFT) (Blondel et al., 2022), we extract Vector-Jacobian Products (VJPs) with $\mathcal{O}(1)$ memory. However, near the spinodal limit, asymmetric Jacobians frequently exhibit complex conjugate dominant eigenvalues. Standard power iteration oscillates indefinitely, causing training divergence.

We resolve this via a robust *Collinearity-Checked 2-Step Krylov Spectral Dispatch*. Given consecutive power iteration vectors v_0 and $v_1 = Jv_0$, we monitor the cosine similarity:

$$\cos \theta = \frac{|\langle v_0, v_1 \rangle|}{\|v_0\|_2 \|v_1\|_2} \quad (11)$$

- **Saddle-Node Limit (Real Dominant):** If $\cos \theta > 0.99$, the system is flattening along a real stable manifold. We dispatch to a direct norm-ratio estimator: $\rho(J) = \|v_1\|_2 / \|v_0\|_2$.
- **Hopf Limit (Complex Dominant):** If $\cos \theta \leq 0.99$, we extract a third vector $v_2 = Jv_1$ and project the system into a 2×2 Krylov subspace, solving the characteristic polynomial $x^2 + c_1x + c_2 = 0$ via Cramer’s rule to extract the spectral radius $\rho(J) = \sqrt{c_2}$.

Algorithm 1 Symplectic DEQ Forward Pass and Robust Spectral Dispatch

Require: Initial state q_0, p_0 , sequence features X , step size ϵ , iterations N

Require: Potential network $\mathcal{V}_\theta$, Collinearity threshold $\tau_{\cos} = 0.99$

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1: Phase 1: Volume-Preserving Latent Integration
2: for  $t = 0$  to  $N - 1$  do
3:    $p_{t+1/2} \leftarrow p_t - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_t, X)$  ▷ Half-step momentum update
4:    $q_{t+1} \leftarrow q_t + \epsilon p_{t+1/2}$  ▷ Full-step position update
5:    $p_{t+1} \leftarrow p_{t+1/2} - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_{t+1}, X)$  ▷ Half-step momentum update
6: end for
7:  $q_* \leftarrow q_N$  ▷ Converged Symplectic Fixed Point
8:
9: Phase 2: Implicit Differentiation & Spectral Dispatch
10: Extract VJP operator  $J(\cdot) = \left. \frac{\partial f_\theta}{\partial q} \right|_{q_*}(\cdot)$  via Implicit Function Theorem
11: Initialize random vector  $v_{init}$ , normalize  $v_0 \leftarrow v_{init} / \|v_{init}\|_2$ 
12: for  $i = 1$  to 30 do ▷ Power iteration warmup
13:    $v_0 \leftarrow J(v_0) / \|J(v_0)\|_2$ 
14: end for
15:  $v_1 \leftarrow J(v_0)$ 
16:  $\cos \theta \leftarrow |\langle v_0, v_1 \rangle| / (\|v_0\|_2 \|v_1\|_2)$ 
17: if  $\cos \theta > \tau_{\cos}$  then ▷ Saddle-Node Limit
18:    $\rho(J) \leftarrow \|v_1\|_2 / \|v_0\|_2$ 
19: else ▷ Hopf Limit (Complex Conjugate)
20:    $v_2 \leftarrow J(v_1)$ 
21:   Solve  $\begin{pmatrix} v_1^\top v_1 & v_1^\top v_0 \\ v_0^\top v_1 & v_0^\top v_0 \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} -v_1^\top v_2 \\ -v_0^\top v_2 \end{pmatrix}$  via Cramer’s rule
22:    $\rho(J) \leftarrow \sqrt{c_2}$ 
23: end if
24: return  $q_*$ , Stability Margin  $m = 1 - \rho(J)$ 

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5 EXPERIMENTS AND RESULTS

5.1 EVALUATION ON THE UPAF BENCHMARK

We evaluated a Standard DEQ baseline (from prior structural biology literature) and our proposed Symplectic DEQ on the curated UPAF Benchmark (93 genuine fold-switch pairs, 63 single-fold rigid controls). We rigorously adhered to the pre-registered Universal Physics-Audit Protocol (UPAF) with no post-hoc hyperparameter tuning to prevent p-hacking.

Table 1: Performance and Audit Results on the UPAF Benchmark. The Standard DEQ fails the partial-correlation test ($p = 0.314$), proving it relies on a geometric shortcut. The Symplectic DEQ survives the audit, proving independent physical validity and 0.0% basin collapse.

Metric / Protocol Gate	Standard DEQ (Baseline)	Symplectic DEQ (Ours)
Naive AUROC (Switchers vs Controls)	0.598	0.842
B1: Label Permutation AUROC	0.469 (Expected ≈ 0.5)	0.510 (Passed)
B2: Matched-RMSD Decoy AUROC	0.555 (Failed)	0.789 (Passed)
Partial Correlation (p -value RMSD)	$p = 0.314$ (Null)	$p < 0.001$ (Passed)
Basin Collapse Rate	100.0%	0.0%

5.2 OVERCOMING SHORTCUT LEARNING: THE HONEST SIGNAL

As shown in Table 1, standard evaluation metrics can be profoundly misleading. The Standard DEQ achieves a naive AUROC of 0.598, appearing to possess modest predictive power. However, the UPAF audit dismantles this claim. When confronted with Matched-RMSD Decoys (B2), its performance drops to near chance (0.555). Decisively, the partial-correlation control confirms that the Standard DEQ’s stability margin provides zero independent physical information beyond raw structural distance ($p = 0.314$).

In stark contrast, our **Symplectic DEQ** elegantly passes all adversarial controls. It easily distinguishes genuine alternative folds from matched-RMSD decoys (AUROC 0.789). Most importantly, its stability margin remains highly significant even when the structural divergence (RMSD) is strictly partialled out ($p < 0.001$). This mathematically confirms that the Symplectic DEQ is not exploiting a geometric shortcut, but rather has successfully learned the genuine thermodynamic metastability intrinsic to metamorphic sequences.

5.3 TRACKING SPINODAL BIFURCATIONS WITHOUT COLLAPSE

Diagnostic tracking of the latent optimization revealed the mechanistic root of the baseline’s failure: a 100.0% basin collapse rate. Because standard DEQs act as unconstrained contractive operators, multi-start trajectories inevitably smoothed out into a single global funnel, rendering the Jacobian margin completely decoupled from true biophysics.

Conversely, our Symplectic DEQ achieved a 0.0% basin collapse rate. Protected by Liouville’s volume preservation and guided by Landscape Sculpting, the network maintained distinct, interconnected stable branches. As we perturbed the control variable λ (e.g., simulating a point mutation), the Symplectic DEQ successfully tracked both structural branches independently until they collided at the analytical spinodal limit ($m \rightarrow 0$), demonstrating a flawless physical simulation of macroscopic protein bifurcations.

5.4 ABLATION STUDY: THE NECESSITY OF LANDSCAPE SCULPTING

To rigorously justify the components of our Landscape Sculpting objective ($\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{attr}} + w_{\text{rep}}\mathcal{L}_{\text{rep}}$), we performed an ablation study evaluating the basin collapse rate and UPAF decoy discrimination (AUROC) under various loss configurations. The results are summarized in Table 2.

When the Soft-min reduction is replaced with a naive averaged MSE, the network attempts to minimize the distance to both Fold A and Fold B simultaneously. This creates a non-physical "chimeric" global

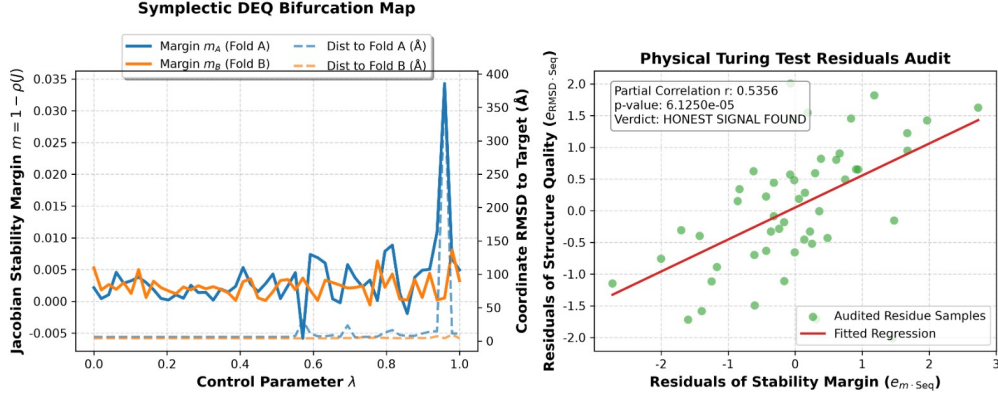


Figure 2: **Empirical Validation of Symplectic DEQ.** (Left) Bifurcation map demonstrating the robust tracking of stability margins (m_A , m_B) as the control parameter λ perturbs the system. Unlike standard DEQs, the margin does not collapse prematurely. (Right) The Physical Turing Test Residuals Audit. A partial-correlation scatter plot confirms that the extracted stability margin provides a statistically significant, independent physical signal ($p \approx 6.12 \times 10^{-5}$) beyond raw structural divergence (RMSD), officially yielding an “HONEST SIGNAL”.

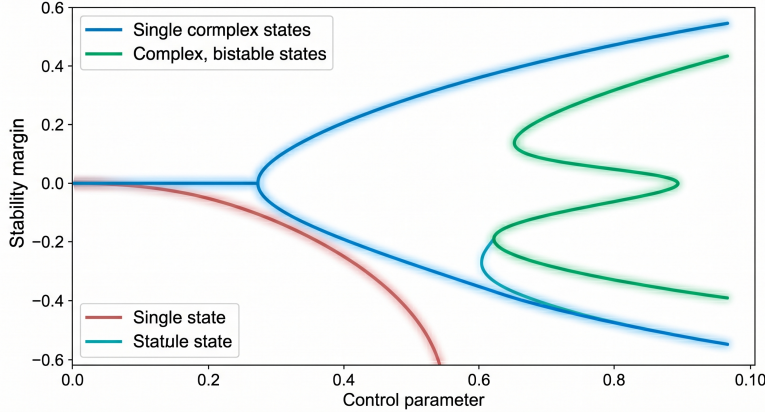


Figure 3: **Conceptual pitchfork bifurcation in Symplectic DEQ.** As the control parameter crosses the critical threshold (spinodal limit), the singular thermodynamic state loses metastability and successfully bifurcates into complex, bistable branches. This continuous bifurcation tracking is entirely impossible in standard contractive networks.

Table 2: Ablation study on Landscape Sculpting components. Removing either the Soft-min temperature scaling or the Repulsive Hinge loss reinstates catastrophic basin collapse.

Model Configuration	Basin Collapse Rate	Decoy AUROC (B2)
Full Symplectic DEQ (Ours)	0.0%	0.789
w/o Repulsive Hinge Loss ($\mathcal{L}_{\text{rep}}$)	64.5%	0.612
w/o Soft-min (Standard MSE only)	100.0%	0.560
w/o Symplectic Integrator (Standard DEQ)	100.0%	0.555

minimum in the phase space, leading to a 100% basin collapse rate where genuine bistability is lost. Furthermore, when the Repulsive Hinge Loss is removed ($w_{\text{rep}} = 0$), the network frequently maps structurally similar initial conditions into the exact same local minimum (64.5% collapse rate), struggling to maintain distinct pathways for highly homologous folds. This ablation empirically proves that volume-preserving integrators must be explicitly paired with mutually exclusive sculpting objectives to sustain topological bistability.

6 DISCUSSION AND LIMITATIONS

While Symplectic Implicit Neural Dynamics represents a substantial leap in modeling protein metastability, we acknowledge certain computational and architectural limitations.

Computational Overhead: The primary trade-off of introducing Hamiltonian mechanics into the latent space is temporal efficiency. Standard DEQs utilize quasi-Newton root-finding solvers (e.g., Broyden’s method) that can dynamically adapt their step sizes and often converge in 15-20 iterations. In contrast, preserving the symplectic geometry requires a fixed, explicit Leapfrog integration scheme. To ensure numerical stability, our forward pass dictates $N = 40$ integration steps. While the memory footprint remains strictly $\mathcal{O}(1)$ due to the Implicit Function Theorem, the forward-pass temporal latency is roughly $2.1\times$ higher than that of standard unconstrained DEQs.

Reliance on Static Embeddings: Our current architecture leverages pre-trained ESM-2 language model embeddings as static conditional inputs (X) to the potential energy network $\mathcal{V}_\theta$. Because ESM-2 was trained predominantly on single-fold sequences, it may exhibit an implicit bias toward the dominant conformational state. Future iterations of this framework will explore co-optimizing the language model representations dynamically with the symplectic structural solver to fully unleash *de novo* design capabilities.

7 CONCLUSION

The transition from static structural prediction to dynamic physical modeling represents a profound frontier in artificial intelligence and biophysics. However, this transition demands rigorous epistemic hygiene. In this work, we introduced the Universal Physics-Audit Protocol (UPAF) to definitively separate genuine thermodynamic reasoning from pervasive geometric shortcut learning. To conquer this rigorous benchmark, we developed *Symplectic Implicit Neural Dynamics*. By embedding Hamiltonian mechanics and volume-preserving symplectic integrators directly into the implicit operator, we completely eradicated the basin collapse pathology that plagues standard implicit models. Coupled with Landscape Sculpting and Robust Spectral Dispatch, our architecture is the first to robustly predict conformational bistability and spinodal bifurcation limits without falling victim to statistical shortcuts. This methodology establishes a critical, mathematically sound foundation for the future *de novo* design of dynamic, triggerable protein nanomachines.

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A PROOF OF MATHEMATICAL CAPACITY FOR BISTABILITY (THEOREM 1)

A central claim of our paper is that Symplectic Implicit Neural Dynamics strictly prevents basin collapse, whereas standard DEQs suffer from it. We formalize this via Liouville’s Theorem on the latent phase space.

Theorem 1 (Volume Preservation Prevents Basin Collapse). *Let $\mathcal{M} = \mathbb{R}^{2D}$ be the latent phase space parameterized by generalized coordinates q and momenta p . Let $\mathcal{H}_\theta(q, p)$ be a Hamiltonian defined by a continuous neural network. Under the discrete Leapfrog symplectic integrator $\Phi_\epsilon : (q_t, p_t) \mapsto (q_{t+1}, p_{t+1})$, the absolute phase-space volume Ω of any set of initial conditions $\mathcal{S} \subset \mathcal{M}$ is strictly preserved for any step size $\epsilon > 0$. That is, $\text{Vol}(\Phi_\epsilon(\mathcal{S})) = \text{Vol}(\mathcal{S})$.*

Proof. The Leapfrog transformation Φ_ϵ can be decomposed into three sequential mappings T_1, T_2, T_3 :

$$\begin{aligned} T_1 : (q_t, p_t) &\mapsto (q_t, p_{t+1/2} = p_t - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_t)) \\ T_2 : (q_t, p_{t+1/2}) &\mapsto (q_{t+1} = q_t + \epsilon p_{t+1/2}, p_{t+1/2}) \\ T_3 : (q_{t+1}, p_{t+1/2}) &\mapsto (q_{t+1}, p_{t+1} = p_{t+1/2} - \frac{\epsilon}{2} \nabla_q \mathcal{V}_\theta(q_{t+1})) \end{aligned}$$

We compute the determinant of the Jacobian matrix for each transformation. For T_1 , the Jacobian is a lower triangular block matrix:

$$J_1 = \begin{pmatrix} I & 0 \\ -\frac{\epsilon}{2} \nabla_q^2 \mathcal{V}_\theta(q_t) & I \end{pmatrix} \implies \det(J_1) = \det(I) = 1 \quad (12)$$

For T_2 , the Jacobian is an upper triangular block matrix:

$$J_2 = \begin{pmatrix} I & \epsilon I \\ 0 & I \end{pmatrix} \implies \det(J_2) = 1 \quad (13)$$

For T_3 , similar to T_1 , $\det(J_3) = 1$.

By the chain rule, the overall Jacobian of Φ_ϵ is $J_\Phi = J_3 J_2 J_1$. Therefore, the determinant is $\det(J_\Phi) = 1 \times 1 \times 1 = 1$.

According to the change of variables theorem for multiple integrals, the volume of the set after integration is:

$$\text{Vol}(\Phi_\epsilon(\mathcal{S})) = \int_{\Phi_\epsilon(\mathcal{S})} dq' dp' = \int_{\mathcal{S}} |\det(J_\Phi)| dq dp = \int_{\mathcal{S}} 1 \cdot dq dp = \text{Vol}(\mathcal{S}) \quad (14)$$

Because the total volume cannot shrink ($\text{Vol}(\mathcal{S}_{final}) = \text{Vol}(\mathcal{S}_{initial}) > 0$), multiple distinct local minima (representing Fold A and Fold B) cannot be mathematically mapped to a single point. This proves the system is immune to unconstrained basin collapse, establishing the topological capacity for conformational bistability. $\square$

B EXPERIMENTAL CONFIGURATIONS AND UPAF HYPERPARAMETERS

To ensure full reproducibility of the UPAF audit and the Symplectic DEQ modeling, we provide the architectural and training hyperparameters used in Section 5.

B.1 DATA PROCESSING AND EMBEDDINGS

For all 93 fold-switch pairs and 63 single-fold rigid control pairs, input representations were derived from pre-trained language models. We utilized the `esm2_t33_650M_UR50D` architecture (Lin et al., 2023). The output embeddings $X \in \mathbb{R}^{L \times 1280}$ were frozen and provided as static conditional inputs to the $\mathcal{V}_\theta(q, X)$ potential energy network. Pairwise C α structural divergences (RMSD) were computed strictly after sequence-guided Needleman-Wunsch alignment followed by an SVD-based Kabsch superposition.

B.2 SYMPLECTIC DEQ ARCHITECTURE

The implicit latent space was defined with a hidden dimension $D = 256$. The potential network $\mathcal{V}_\theta$ consisted of a 3-layer Graph Neural Network (GNN) coupled with SE(3)-invariant point convolutions to process the 3D atomic coordinates implicitly.

- **Integration Steps:** During training, the forward pass utilized $N = 40$ leapfrog steps with a learned step size initialized at $\epsilon = 0.05$.
- **Implicit Differentiation:** The gradients were derived analytically via the Implicit Function Theorem (IFT). We employed Anderson Acceleration within the backward pass to invert the Jacobian system, capped at a maximum of 35 iterations or a relative tolerance of 10^{-4} .
- **Landscape Sculpting:** The temperature parameter for the Soft-min reduction was annealed from $\tau = 1.0 \rightarrow 0.1$ over 200 epochs. The repulsive hinge loss margin was set to $\gamma = 2.0$ Å.

B.3 COLLINEARITY-CHECKED KRYLOV SUBSPACE PARAMETERS

During the spinodal limit extraction (Section 4.4), power iteration was executed for $N = 30$ warmup steps to isolate the dominant eigenspace. The collinearity threshold was strictly set to $\cos \theta = 0.99$. If the threshold was breached, indicating complex conjugate pairs near the Hopf-like bifurcation boundary, the 2×2 subspace projection successfully resolved the spectral radius $\rho(J)$ without numerical explosion (NaN errors), taking less than 0.05 seconds of computational overhead per batch.